Unit no. º 6

FUNCTIONS: REPRESENTATIONS AND REFERENCE FUNCTIONS REFERENCE

23 November-4 December 2026

I Function, image and preimage

Definition

A real-valued function assigns to each real number x of its domain of definition D_f a unique real $f(x)$, called the image of x. A preimage of y is a real $x \in D_f$ such that $f(x) = y$. We write $f : x \mapsto f(x)$. A number may have no, one or several preimages.

Example 6.1

Let $g(x) = x^2 - 4x + 32$ on $\mathbb{R}$. Calculate $g(-5)$. Let $h(x) = 3x + 5$; find the preimage of 6 under h.

Solution:

$g(-5) = 25 + 20 + 32 = 77$: 77 is the image of -5, and -5 is a preimage of 77. For h, $3x + 5 = 6 \iff x = 1/3$. The only preimage is $1/3$.

Example 6.2

For the square function, calculate the images of -3 and 3, then the preimages of 9, 0 and -1.

Solution:

$f(-3) = f(3) = 9$. The preimages of 9 are -3, 3; 0 has a single preimage, 0; -1 has none in $\mathbb{R}$.

```
def g(x):
    return x*x - 4*x + 32
```

Example 6.3

Distinguish `g(-5)` vs `x = -5`.

Solution:

The first calls the function and returns 77. The second assigns -5 to the variable `x`, without calculating the image under g.

II Domain

Conditions for existence

D_f contains the values for which the process defining f makes sense. A denominator must be non-zero; a number under a square root must be positive or zero. Also respect any domain imposed by the question or the context.

Example 6.4

Determine the domains:

$$f(x) = 3x^2 - 5x + 2, \quad g(x) = \frac{2x+1}{x-4}, \quad h(x) = \frac{x}{x^2-9}, \quad k(x) = \sqrt{2x-6}.$$

Solution:

$D_f = \mathbb{R}$. $D_g =]-\infty; 4[\cup]4; +\infty[$ as $x - 4 \neq 0$. $D_h =]-\infty; -3[\cup]-3; 3[\cup]3; +\infty[$ as $x^2 - 9 \neq 0$. $D_k = [3; +\infty[$ since $2x - 6 \geq 0$.

Watch out. The square root of a negative number does not exist in $\mathbb{R}$; it is not "a negative square root" that is being excluded.

Example 6.5: Modelling

A rectangle with perimeter 20 m has width x m. Express its area and state the domain if both dimensions must be strictly positive.

Solution:

The length is $10 - x$, therefore $A(x) = x(10 - x)$ in m ^{2}. $x > 0$ and $10 - x > 0$ give $D_A =]0; 10[$. If degenerate rectangles were allowed, we could take $[0; 10]$.

III Table of values and graph

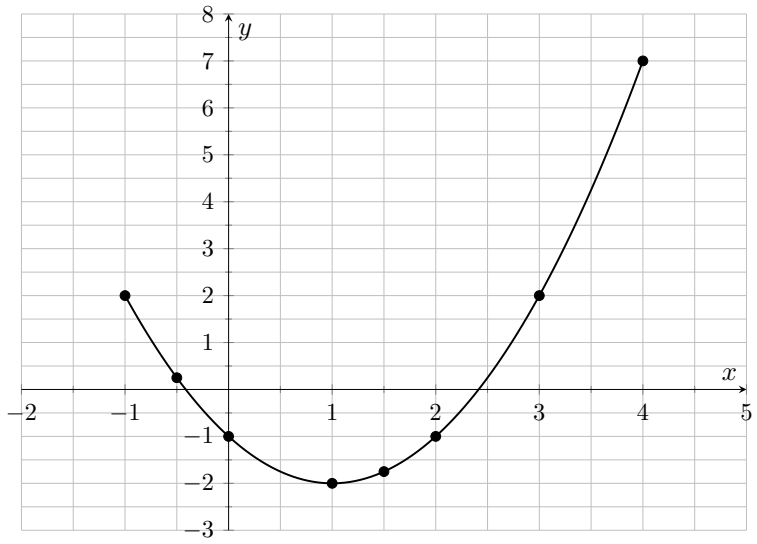
Definition

$\mathcal{C}_f$ is the set of points ($x; f(x)$), with $x \in D_f$. Thus $M(a;b) \in \mathcal{C}_f \iff a \in D_f$ and $b = f(a)$.

Example 6.6

On $[-1;4]$, $f(x) = x^2 - 2x - 1$. Complete the table, then plot the curve over the entire interval.

x	-1	$-0{,}5$	0	1	$1{,}5$	2	3	4
$f(x)$	2	$0{,}25$	-1	-2	$-1{,}75$	-1	2	7



Solution:

Images: 2; 0 ,25 ; -1 ; -2 ; $-1{,}75$; -1; 2; 7. Plot the points then draw the parabola. Segments linking these points alone would only give an approximation.

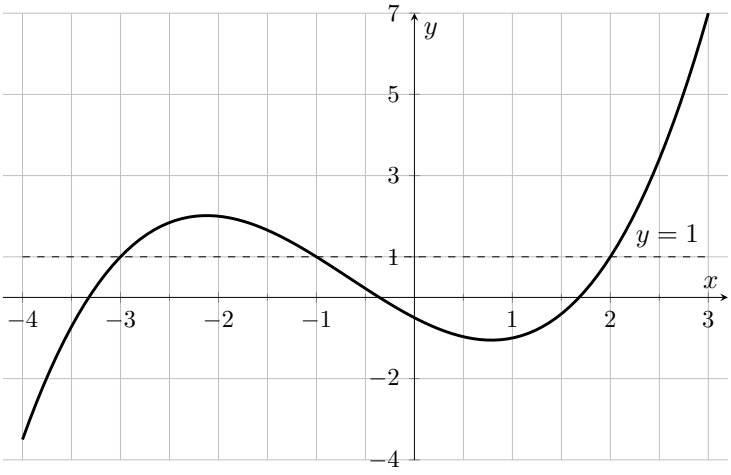
IV Solving equations and inequalities graphically

Method

The solutions of $f(x) = k$ are the x-coordinates of the intersections of $\mathcal{C}_f$ with $y = k$. For $f(x) \leq k$, read off the x-coordinates where the curve is below this line, intersections included; with $<$, the intersections are excluded.

Example 6.7

The curve represents $f(x) = 1 + \frac{1}{4}(x+3)(x+1)(x-2)$ on $[-4; 3]$. Solve graphically $f(x) = 1$, $f(x) \leq 1$ and $f(x) > 1$.



Solution:

$f(x) = 1 : -3, -1, 2$. $f(x) \leq 1 : [-4; -3] \cup [-1; 2]$. $f(x) > 1 :]-3; -1[\cup]2; 3]$. The endpoints $-4, 3$ come from the domain. Factorisation checks exactly the intersections; the algebraic study of the sign will be consolidated in unit 8.

Guidance for the teacher

Graph redrawn to replace an ambiguous reading in the original booklet. Distinguish graphical conjecture and exact verification.

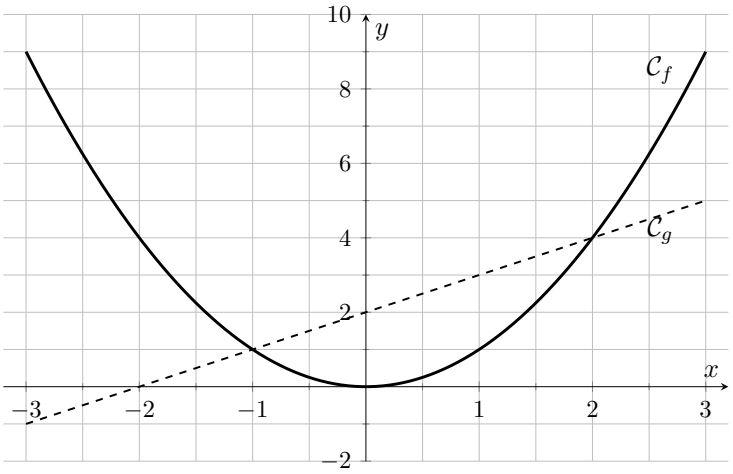
V Compare two functions; read a sign

Method

$f(x) = g(x)$ is read at the intersections of the curves. For $f(x) < g(x)$, $\mathcal{C}_f$ is strictly below of $\mathcal{C}_g$ at the same x-coordinate. The sign of $f(x)$ is read with respect to the x-axis.

Example 6.8

On $[-3;3]$, $f(x) = x^2$ & $g(x) = x + 2$. Solve $f = g$, $f < g$, $f \geq g$ and determine the sign of g.



Solution:

$f = g$ for $x = -1, 2$; $f < g$ on $]-1;2[$; $f \geq g$ on $[-3;-1] \cup [2;3]$. g is strictly negative on $[-3;-2[$, zero at -2, strictly positive on $]-2;3]$. Sign and variation are different facts.

VI Affine functions and absolute value

Definitions

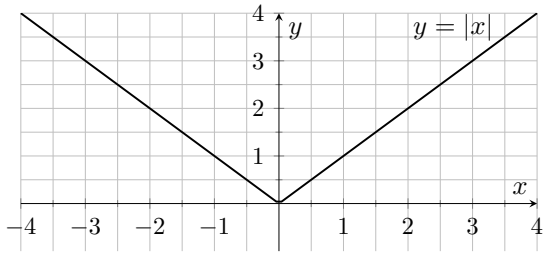
An affine function is $f(x) = mx + p$ on $\mathbb{R}$. It is linear if $p = 0$, constant if $m = 0$. For $m \neq 0$, it vanishes at $-p/m$; its sign is that of m to the right of this number. The function $x \mapsto |x|$ is defined on $\mathbb{R}$, positive or zero, and vanishes only at 0.

Example 6.9

Classify (several answers possible) as affine, linear, constant: 5 $x + 2$; $5x$; $7 + 2x - 7$; 6 ; $6(4x - 2)$; $6x + 5 - 6x$.

Solution:

All are affine. 5 x and $2x$ are linear; 6 and 5 are constant; 24 $x - 12$ is neither linear nor constant.



Equations and inequalities

If $k < 0$, $|x| = k$ and $|x| \leq k$ have no solution, and $|x| \geq k$ is true on $\mathbb{R}$. If $k = 0$, the first two give $x = 0$, the third is true on $\mathbb{R}$. For $k > 0$:

$$|x| = k \iff x \in \{-k; k\}, \quad |x| \leq k \iff x \in [-k; k],$$

$$|x| \geq k \iff x \in]-\infty; -k] \cup [k; +\infty[.$$

Example 6.10

Solve $|x| = 5$, $|x| = -3$, $|x| \leq 3$, $|x| \geq 4$.

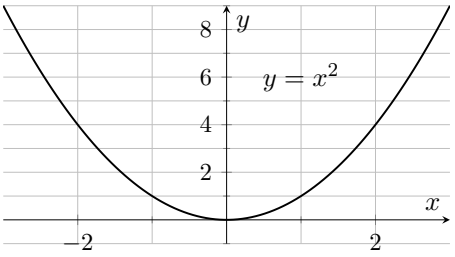
Solution:

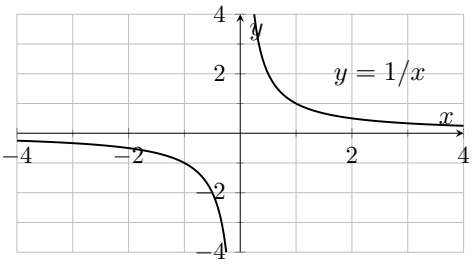
$\{-5; 5\}$; $\varnothing$; $[-3; 3]$; $]-\infty; -4] \cup [4; +\infty[$.

VII Square and reciprocal functions

Definitions and sign

$x \mapsto x^2$ is defined on $\mathbb{R}$; its graph is a parabola. $x^2 \geq 0$, with equality only at 0. $x \mapsto 1/x$ is defined on $\mathbb{R}^*$; its graph is a hyperbola. 1 $/x$ has the sign of x and is never zero.





Equations and inequalities with x^2

If $k < 0$, $x^2 = k$ and $x^2 \leq k$ have no solution; $x^2 \geq k$ is true on $\mathbb{R}$. If $k = 0$, the first two give $x = 0$, the third is true on $\mathbb{R}$. For $k > 0$:

$$x^2 = k \iff x \in \{-\sqrt{k}; \sqrt{k}\}, \quad x^2 \leq k \iff x \in [-\sqrt{k}; \sqrt{k}],$$

$$x^2 \geq k \iff x \in]-\infty; -\sqrt{k}] \cup [\sqrt{k}; +\infty[.$$

Example 6.11

Solve $x^2 = 4$, $x^2 = -1$, $x^2 \geq 4$, $x^2 \geq -1$, $1/x = 4$, $1/x = 0$.

Solution:

$\{-2; 2\}$; $\varnothing$; $]-\infty; -2] \cup [2; +\infty[$; $\mathbb{R}$; $\{1/4\}$; $\varnothing$. More generally, 1 $/x = k$ has the unique solution $1/k$ if $k \neq 0$, none if $k = 0$.

VIII Inequalities with the reciprocal function

Method

Remember $x \neq 0$ and read both branches of the hyperbola. Do not multiply an inequality by x without knowing its sign.

	$k < 0$	$k = 0$	$k > 0$
$1/x \leq k$	$[1/k; 0[$	$]-\infty; 0[$	$]-\infty; 0[\cup[1/k; +\infty[$
$1/x \geq k$	$]-\infty; 1/k]\cup]0; +\infty[$	$]0; +\infty[$	$]0; 1/k]$

Example 6.12

Solve 1 $/x \geq 2$, $1/x \leq -1$, $1/x < 1/2$ and $1/x > -2$.

Solution:

$]0; 1/2]$; $[-1; 0[$; $]-\infty; 0[\cup]2; +\infty[$; $]-\infty; -1/2[\cup]0; +\infty[$. A strict inequality excludes the intersection with $y = k$; 0 is always forbidden.

Example 6.13

A student writes: "1 $/x \leq 2 \iff 1 \leq 2x$, therefore $x \geq 1/2$". What is missing?

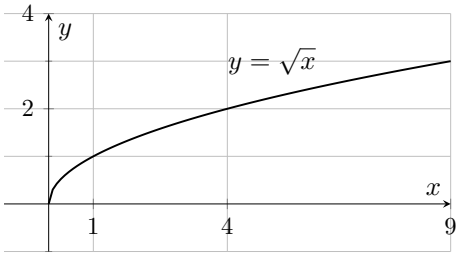
Solution:

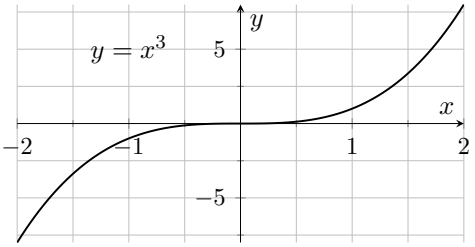
The student implicitly assumed $x > 0$. For $x < 0$, $1/x < 0 < 2$: these values also work. The correct set is $]-\infty; 0[\cup[1/2; +\infty[$. The number -1 refutes his equivalence.

IX Using square roots and cubes

Definitions

$x \mapsto \sqrt{x}$ is defined on [0; + ∞[and takes positive or zero values. For $k \geq 0$, $\sqrt{x} = k \iff x = k^2$; for $k < 0$, no solution. $x \mapsto x^3$ is defined on $\mathbb{R}$. Each real number k has a unique preimage, denoted $\sqrt[3]{k}$.





Example 6.14

Solve $\sqrt{x} = 3$; $\sqrt{x} = -2$; $\sqrt{x} \leq 2$; $\sqrt{x} > 3$; $x^3 = -8$; $x^3 \leq 27$.

Solution:

$\{9\}$; $\varnothing$; $[0; 4]$; $]9; +\infty[$; $\{-2\}$; $]-\infty; 3]$. Respect the domain for square roots. The cube function preserves order.

Example 6.15

Compare x^2 & x for $x = 1/2$, then x = 2. Is the square of a positive number always greater than it?

Solution:

$1/4 < 1/2$ but 4 > 2. The statement is false; 1 /2 is a counterexample. General study in unit 10.